Thrissur, Kerala
aaron123thomas098@gmail.com
arxhr007.github.io
linkedin.com/in/aaron-thomas-parakkal
github.com/arxhr007

Aaron Thomas

Profile

Final year Computer Science student building production backends. REST APIs in Python/FastAPI and Node.js, PostgreSQL and Firebase data models, and AWS deployments, from government CMS and LMS platforms to Android app backends.

Technical Skills

- **Languages:** Python, TypeScript/JavaScript, Rust, Java, C
- **Backend:** Node.js, FastAPI, REST APIs, PostgreSQL, MongoDB, Firebase, Supabase
- **Frontend:** React, Next.js, React Native, Tailwind CSS
- **Cloud/DevOps:** AWS, GCP, Docker, CI/CD
- **AI/ML:** LangChain, LangGraph, LlamaIndex, Prompt Engineering
- **Soft Skills:** Communication, Teamwork, Leadership, Problem Solving

Experience

Backend Developer (Contract) — **Dubai, UAE (Remote)**
Lions Information Technology L.L.C — *04/2026 – 08/2026*
- Built backend services and REST APIs powering Android applications using Python and FastAPI.
- Designed PostgreSQL data models and deployed services on AWS for mobile clients.

Software Engineering Intern — **Thiruvananthapuram, KL**
Stride, K-DISC — *10/2025 – Present*
- Developed enterprise CMS and LMS platforms for the Government of Kerala using Next.js, Node.js, and AWS.
- Built scalable backend services, REST APIs, and cloud-integrated workflows for reliable content delivery.

Backend & Automation Engineer (Part-time) — **Kodakara, KL**
Sahrdaya College of Engineering & Technology (Autonomous) — *08/2025 – 02/2026*
- Architected backend systems for the official college website and CMS using Firebase, Node.js, and REST APIs.
- Built workflow automation and system orchestration to streamline CMS operations and reduce repetitive manual work.

Web Development Intern — **New Zealand (Remote)**
Pexcel Solutions — *12/2024 – 06/2025*
- Developed responsive frontend interfaces using React.js and Tailwind CSS for international client projects.
- Integrated SEO best practices that increased organic search visibility for project deployments.

Projects

Aliens Eye — github.com/arxhr007/Aliens_eye
AI-Powered Public Data Analysis Platform
- Built an AI-powered platform for large-scale public data analysis, with parallel processing pipelines.
- Achieved **4000 GitHub stars**.

AYRA — blog.myosa-sensors.org/blog.html?file=blogs/ayra.md
TinyML-Based Autonomic Dysreflexia Early Warning System
- Developed a below-knee wearable for real-time autonomic dysreflexia early warning, fusing **GSR, PPG/HRV, ECG, skin temperature, humidity, and motion** signals from a single placement.
- Implemented an **int8 quantized TinyML neural network** on an ESP32.
- Presented at the **IEEE BioSensors 2026** conference in Bangkok, Thailand.

NeuroLearn — github.com/arxhr007/neurolearn
Adaptive AI Tutor
- Built an adaptive tutor for neurodivergent learners using LangGraph, personalized learning flows, and RAG.
- Presented at an OpenAI Community Meetup.

Amma — github.com/arxhr007/amma
Malayalam Programming Language
- Designed a custom interpreter in Rust with a lexer, parser, and evaluator supporting control flow, recursion, and file I/O.
- Built a transliteration layer so code runs in Malayalam script or romanized English.

Leadership

- **Coders Club Lead, IEDC Sahrdaya:** Organized developer workshops, project sessions, and innovation activities.
- **Technical Team, IEEE Sahrdaya:** Built event websites, attendance systems, and event management applications.
- **Mentor, TinkerHub:** Guided peers on project architecture, development workflows, and implementation strategy.

Education

B.Tech in Computer Science and Engineering — **Kodakara, KL**
APJ Abdul Kalam Technological University (KTU) | ***CGPA: 6.53/10*** — *2023 – 2027*

Achievements

- **MYOSA International Hackathon Finalist:** Finalist for AYRA, a TinyML wearable for autonomic dysreflexia early warning.
- **IEEE BioSensors 2026:** Presented AYRA at the IEEE BioSensors 2026 international conference in Bangkok, Thailand.
- **4x CTF Winner:** Won four Capture The Flag cybersecurity competitions.